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Inventor(s)

Conrad; Wayne Ernest

SURFACE CLEANING APPARATUS

Abstract

A hand vacuum cleaner has an air flow path from a dirty air inlet to a clean air outlet. A suction motor is positioned in the air flow path and a non-cyclonic air treatment chamber is in the air flow path downstream from the dirty air inlet and upstream from the suction motor. The air treatment chamber has a front end having a chamber air inlet, a rear end having a chamber air outlet. A non-return valve flap closes an inlet port of the chamber air inlet. The non-return valve flap has an upper end that is mounted to the air treatment chamber and a lower end. The non-return valve flap is moveable from a closed position in which it closes the inlet port to an open position in which the lower end of the non-return valve flap rotates inwardly and upwardly whereby the inlet port is opened.

Inventors: Conrad; Wayne Ernest (Hampton, CA)

Applicant: Omachron Intellectual Property Inc. (Hampton, CA)

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Background/Summary

FIELD

[0001] This disclosure relates generally to surface cleaning apparatus. In a preferred embodiment, the surface cleaning apparatus comprises a portable surface cleaning apparatus, such as a hand vacuum cleaner.

INTRODUCTION

[0002] The following is not an admission that anything discussed below is part of the prior art or part of the common general knowledge of a person skilled in the art.

[0003] Various types of surface cleaning apparatus are known, including upright surface cleaning apparatus, canister surface cleaning apparatus, stick surface cleaning apparatus, central vacuum systems, and hand carriable surface cleaning apparatus such as hand vacuums. Further, various designs for cyclonic and non-cyclonic hand vacuum cleaners, including battery operated cyclonic and non-cyclonic hand vacuum cleaners, are known in the art.

SUMMARY

[0004] This summary is intended to introduce the reader to the more detailed description that follows and not to limit or define any claimed or as yet unclaimed invention. One or more inventions may reside in any combination or sub-combination of the elements or process steps disclosed in any part of this document including its claims and figures.

[0005] In accordance with one aspect of this disclosure, which may be used alone or in combination with any other aspect, a surface cleaning apparatus has an air treatment chamber, which has an inlet end and an outlet end, which is spaced in an axial direction from the inlet end. A sidewall extends from the inlet end to the outlet end and the inlet port has a non-return valve, which may be closed when the surface cleaning apparatus is turned off.

[0006] The non-return valve is used to direct air in a direction generally transverse to the axial direction across part or essentially all of the air treatment chamber to or towards the sidewall on a side of the air treatment chamber opposed to the port having the non-return valve. Unlike a tangential air inlet for a cyclone that directs air to flow around a central axis of rotation a cyclone chamber as the air travel longitudinally in a direction of the cyclone axis, this deflection of the air creates a non-cyclonic flow of air in the chamber which may flow axially to a porous separator (e.g., screen) at the outlet end and travel past or along part or all of the porous separator to inhibit particulate matter building up on the porous separator.

[0007] In accordance with this aspect, there is provided a hand vacuum cleaner has

[0008] an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet. A suction motor is positioned in the air flow path. A handle is provided on a lower side of the hand vacuum cleaner. A non-cyclonic air treatment chamber is in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet, and a longitudinal axis extending between the front and rear ends of the air treatment chamber. A non-return valve flap closes an inlet port of the chamber air inlet, the non-return valve flap having an upper end that is mounted to the air treatment chamber and a lower end. The non-return valve flap is moveable from a closed position in which it closes the inlet port to an open position in which the lower end of the non-return valve flap rotates inwardly and upwardly whereby the inlet port is opened.

[0009] In accordance with this aspect, there is also provided a hand vacuum cleaner has an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet. A suction motor is positioned in the air flow path. The dirty air inlet is provided at an upper end of the hand vacuum cleaner. A non-cyclonic air treatment chamber is in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet, and a longitudinal axis extending between the front and rear ends of

the air treatment chamber. A non-return valve flap closes an inlet port of the chamber air inlet, the non-return valve flap having an upper end that is mounted to the air treatment chamber and a lower end. The non-return valve flap is moveable from a closed position in which it closes the inlet port to an open position in which the lower end of the non-return valve flap rotates inwardly and upwardly whereby the inlet port is opened.

[0010] In accordance with this aspect, there is also provided a hand vacuum cleaner has an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet. A suction motor is positioned in the air flow path. A non-cyclonic air treatment chamber is in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet, a sidewall extending between the front and rear ends of the air treatment chamber, and a longitudinal axis extending between the front and rear ends of the air treatment chamber. A non-return valve flap closes an inlet port of the chamber air inlet, the non-return valve flap having a first end that is mounted to the air treatment chamber at a location adjacent a first portion of an inner surface of the sidewall and an opposed end. The non-return valve flap is moveable from a closed position in which it closes the inlet port to an open position in which the opposed end of the non-return valve flap rotates inwardly and towards the sidewall whereby the inlet port is opened and air exiting the inlet port is directed across a volume of the non-cyclonic air treatment chamber towards a second portion of the sidewall that is opposed to the first portion in a direction transverse to the longitudinal axis.

[0011] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, a surface cleaning apparatus has an air treatment chamber, which has an inlet end and an outlet end, which is spaced in an axial direction from the inlet end. A sidewall extends from the inlet end to the outlet end and a baffle plate is located axially inward of an inlet port of the air treatment chamber. Accordingly, in operation, some or all of the air exiting the inlet port and entering the air treatment chamber may impinge upon the baffle plate and be inhibited or prevented from traveling directly axially towards a porous separator (e.g., screen) provided at the outlet end.

[0012] In accordance with this aspect, there is provided a hand vacuum cleaner has an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet. A suction motor is positioned in the air flow path. A non-cyclonic air treatment chamber is in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet, a sidewall extending between the front and rear ends of the air treatment chamber, and a longitudinal axis extending between the front and rear ends of the air treatment chamber. The chamber air inlet has an inlet axis that extends rearwardly. A baffle plate is positioned in the air treatment chamber rearward of the chamber air inlet and the inlet axis intersects the baffle plate.

[0013] In accordance with this aspect, there is also provided a hand vacuum cleaner has an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet. A suction motor is positioned in the air flow path. A non-cyclonic air treatment chamber is in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet, and a longitudinal axis extending between the front and rear ends of the air treatment chamber. A baffle plate is positioned in the air treatment chamber rearward of the chamber air inlet overlying an outlet port of the chamber air inlet.

[0014] In accordance with another aspect of this disclosure, which may be used alone or in combination with any other aspect, a surface cleaning apparatus has an air treatment chamber, which has an inlet end and an outlet end, which is spaced in an axial direction from the inlet end. A sidewall extends from the inlet end to the outlet end and an air impervious cover is positioned at the

outlet end of the air treatment chamber so as to be positioned axially between the inlet port and a porous outlet member (e.g., screen). Accordingly, the air impervious cover may function as a barrier wall to inhibit or prevent air entering the air treatment chamber through the inlet port from travelling directly to the porous outlet member. Air which travels directly axially to the porous outlet member may have a sufficient velocity and/or angle of impact with the porous outlet member that particulate matter may be pushed through or embedded in openings of the porous outlet member.

[0015] In accordance with this aspect, there is provided a hand vacuum cleaner has an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet. A suction motor is positioned in the air flow path. A non-cyclonic air treatment chamber is in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet, a sidewall extending between the front and rear ends of the air treatment chamber, and a longitudinal axis extending between the front and rear ends of the air treatment chamber. The chamber air outlet includes a porous separator. An air impervious cover is positioned at the rear end of the air treatment chamber and overlying the porous separator whereby an air flow path through the air treatment chamber includes a portion that extends generally radially inwardly at a location rearward of the air impervious cover.

[0016] These and other aspects and features of various embodiments will be described in greater detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] For a better understanding of the described embodiments and to show more clearly how they may be carried into effect, reference will now be made, by way of example, to the accompanying drawings in which:

[0018] FIG. 1 is a perspective view of an example hand vacuum cleaner attached to a wand and surface cleaning head;

[0019] FIG. 2 is a perspective view of the hand vacuum cleaner of FIG. 1 removed from the wand and surface cleaning head;

[0020] FIG. 3 is a side view of the hand vacuum cleaner of FIG. 1;

[0021] FIG. 4 is a top view of the hand vacuum cleaner of FIG. 1;

[0022] FIG. 5 is a front view of the hand vacuum cleaner of FIG. 1;

[0023] FIG. 6 is a side cross-sectional view taken through the lateral centre of the hand vacuum cleaner of FIG. 1;

[0024] FIG. 7 is a side view of the hand vacuum cleaner of FIG. 1 with an openable portion in an open position;

[0025] FIG. 8 is a perspective view of the hand vacuum cleaner of FIG. 1 with the openable portion in the open position;

[0026] FIG. 9A is a perspective view of the hand vacuum cleaner of FIG. 1 docked at a docking station;

[0027] FIG. 9B is an enlarged vertical cross-sectional view of the hand vacuum cleaner of FIG. 1 when docked at the docking station and having an openable door in a closed position;

[0028] FIG. 9C is an enlarged vertical cross-sectional view of the hand vacuum cleaner of FIG. 1 when docked at the docking station during an evacuation operation with an openable door in an open position;

[0029] FIG. 10A is a perspective view of an example dome-shaped porous separator for use with the hand vacuum cleaner of FIG. 1;

[0030] FIG. 10B is a top perspective partially cut away view of the hand vacuum cleaner of FIG. 1 having the dome-shaped porous separator of FIG. 10A;

[0031] FIG. 10C is a side perspective cross-sectional view of the hand vacuum cleaner of FIG. 1 having the dome-shaped porous separator of FIG. 10A;

[0032] FIG. 11A is a perspective view of an example flat porous separator for use with the hand vacuum cleaner of FIG. 1;

[0033] FIG. 11B is a top perspective partially cut away view of the hand vacuum cleaner of FIG. 1 having the flat porous separator of FIG. 11A;

[0034] FIG. 11C is a side perspective cross-sectional view of the hand vacuum cleaner of FIG. 1 having the flat porous separator of FIG. 11A;

[0035] FIG. 12A is a perspective view of an example conical porous separator for use with the hand vacuum cleaner of FIG. 1;

[0036] FIG. 12B is a top perspective partially cut away view of the hand vacuum cleaner of FIG. 1 having the conical porous separator;

[0037] FIG. 12C is a perspective cross-sectional view of the hand vacuum cleaner of FIG. 1 having the conical porous separator having the conical porous separator of FIG. 12A;

[0038] FIG. 13A is a side cross-sectional view of the hand vacuum cleaner of FIG. 1 having a non-return valve in a closed position having the conical porous separator of FIG. 12A;

[0039] FIG. 13B is a side cross-sectional view of the hand vacuum cleaner of FIG. 1 having the non-return valve in an open position;

[0040] FIG. 14 is a front cross-sectional view of the hand vacuum cleaner of FIG. 1 having the non-return valve in the closed position;

[0041] FIG. 15 is a top cross-sectional view of the hand vacuum cleaner of FIG. 1 having the non-return valve in the open position;

[0042] FIG. 16A is a front cross-sectional view of the hand vacuum cleaner of FIG. 1 with a baffle plate that has a planar free edge;

[0043] FIG. 16B is a front cross-sectional view of the hand vacuum cleaner of FIG. 1 with a baffle plate that has two recesses in the free edge;

[0044] FIG. 16C is a front cross-sectional view of the hand vacuum cleaner of FIG. 1 with a baffle plate that has a central protrusion extending from the free edge;

[0045] FIG. 17A is a top cross-sectional view of the hand vacuum cleaner of FIG. 1 with a baffle plate that has a protrusion extending from a front side;

[0046] FIG. 17B is a perspective cross-sectional view of the hand vacuum cleaner of FIG. 1 with a baffle plate that has a protrusion extending from the front side;

[0047] FIG. 18A is a side cross-sectional view of the hand vacuum cleaner of FIG. 1 having a baffle plate with a planar rear side;

[0048] FIG. 18B is a side cross-sectional view of the hand vacuum cleaner of FIG. 1 having a baffle plate with a curved rear side;

[0049] FIG. 18C is a side cross-sectional view of the hand vacuum cleaner of FIG. 1 having a baffle plate with an angled rear side;

[0050] FIG. 19A is a side cross-sectional view of the hand vacuum cleaner of FIG. 1 having an axially extending rib along a sidewall of an air treatment chamber;

[0051] FIG. 19B is a side cross-sectional view of the hand vacuum cleaner of FIG. 1 having the axially extending rib along a portion of the sidewall of the air treatment chamber;

[0052] FIG. 20A is a top perspective partially cut away view of the hand vacuum cleaner of FIG. 1 having an air impervious cover;

[0053] FIG. 20B is a side perspective cross-sectional view of the hand vacuum cleaner of FIG. 20A showing the air impervious cover overlying a vertical porous separator;

[0054] FIG. 20C is a side cross-sectional view of the hand vacuum cleaner of FIG. 20A having the air impervious cover overlying the vertical porous separator;

[0055] FIG. 20D is a perspective cross-sectional view of the hand vacuum cleaner of FIG. 20A having the air impervious cover overlying a horizontal porous separator;

[0056] FIG. 20E is a cross-sectional view of the hand vacuum cleaner of FIG. 20A having the air impervious cover overlying the horizontal porous separator;

[0057] FIG. 21A is a top perspective partially cut away view of the hand vacuum cleaner of FIG. 1 having an alternate air impervious cover;

[0058] FIG. 21B is a perspective cross-sectional view of the hand vacuum cleaner of FIG. 21A with the air impervious cover overlying another horizontal porous separator; and,

[0059] FIG. 21C is a side cross-sectional view of the hand vacuum cleaner of FIG. 21A with the air impervious cover overlying the horizontal porous separator.

[0060] The drawings included herewith are for illustrating various examples of articles, methods, and apparatuses of the teaching of the present specification and are not intended to limit the scope of what is taught in any way.

DESCRIPTION OF VARIOUS EMBODIMENTS

[0061] Various apparatus, methods and compositions are described below to provide an example of an embodiment of each claimed invention. No embodiment described below limits any claimed invention and any claimed invention may cover apparatuses and methods that differ from those described below. The claimed inventions are not limited to apparatus, methods and compositions having all of the features of any one apparatus, method or composition described below or to features common to multiple or all of the apparatus, methods or compositions described below. It is possible that an apparatus, method or composition described below is not an embodiment of any claimed invention. Any invention disclosed in an apparatus, method or composition described below that is not claimed in this document may be the subject matter of another protective instrument, for example, a continuing patent application, and the applicant(s), inventor(s) and/or owner(s) do not intend to abandon, disclaim, or dedicate to the public any such invention by its disclosure in this document.

[0062] The terms “an embodiment”, “embodiment”, “embodiments”, “the embodiment”, “the embodiments”, “one or more embodiments,” “some embodiments”, and “one embodiment” mean “one or more (but not all) embodiments of the present invention(s)”, unless expressly specified otherwise.

[0063] The terms “including”, “comprising”, and variations thereof mean “including but not limited to”, unless expressly specified otherwise. A listing of items does not imply that any or all of the items are mutually exclusive, unless expressly specified otherwise. The terms “a”, “an”, and “the” mean “one or more”, unless expressly specified otherwise.

[0064] As used herein and in the claims, two or more parts are said to be “coupled”, “connected”, “attached”, or “fastened” where the parts are joined or operate together either directly or indirectly (i.e., through one or more intermediate parts), so long as a link occurs. As used herein and in the claims, two or more parts are said to be “directly coupled”, “directly connected”, “directly attached”, or “directly fastened” where the parts are connected in physical contact with each other. None of the terms “coupled”, “connected”, “attached”, and “fastened” distinguish the manner in which two or more parts are joined together.

[0065] Furthermore, it will be appreciated that for simplicity and clarity of illustration, where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the example embodiments described herein. However, it will be understood by those of ordinary skill in the art that the example embodiments described herein may be practiced without these specific details. In other instances, well-known methods, procedures, and components have not been described in detail so as not to obscure the example embodiments described herein. Also, the description is not to be considered as limiting the scope of the example embodiments described herein.

[0066] As used herein, the wording “and/or” is intended to represent an inclusive-or. That is, “X and/or Y” is intended to mean X or Y or both, for example. As a further example, “X, Y, and/or Z” is intended to mean X or Y or Z or any combination thereof.

[0067] As used herein and in the claims, two elements are said to be “parallel” where those elements are parallel and spaced apart, or where those elements are collinear.

[0068] Some elements herein may be identified by a part number, which is composed of a base number followed by an alphabetical or subscript-numerical suffix (e.g., **300a**, or **3001**). Multiple elements herein may be identified by part numbers that share a base number in common and that differ by their suffixes (e.g., **3001**, **3002**, and **3003**). All elements with a common base number may be referred to collectively or generically using the base number without a suffix (e.g., **300**).

[0069] It should be noted that terms of degree such as “substantially”, “about”, and “approximately” as used herein mean a reasonable amount of deviation of the modified term such that the end result is not significantly changed. These terms of degree may also be construed as including a deviation of the modified term, such as by 1%, 2%, 5% or 10%, for example, if this deviation does not negate the meaning of the term it modifies. For example, the expressions “substantially perpendicular” and “substantially parallel” mean within 10% of perpendicular and parallel, respectively.

General Description of a Hand Vacuum Cleaner

[0070] FIGS. **1** to **9** show an example embodiment of a surface cleaning apparatus **100**. The following is a general discussion of the surface cleaning apparatus **100** which provides a basis for understanding several of the features that are discussed herein. As discussed in detail subsequently, each of the features may be used in other embodiments.

[0071] In the embodiments illustrated, the surface cleaning apparatus **100** is a hand-held vacuum cleaner, which is commonly referred to as a “hand vacuum cleaner” or a “handvac”. As used herein, a hand-held vacuum cleaner or hand vacuum cleaner or handvac is a vacuum cleaner that can be operated generally one-handedly to clean a surface while its weight is held by the same one hand. This is contrasted with upright and canister vacuum cleaners, the weight of which is supported by a surface (e.g., floor below) during use.

[0072] In alternate embodiments, the features of the air treatment chamber discussed herein may be used in any other type of surface cleaning apparatus such as an upright surface cleaning apparatus, a canister surface cleaning apparatus, wet-dry vacuum cleaner, an extractor or the like.

[0073] Optionally, as exemplified in FIG. **1**, the hand vacuum **100** can be removably mounted to a base, such as base **102** (either removably or in a fixed manner) so as to form, for example, an upright vacuum cleaner, a canister vacuum cleaner, a stick vac, a wet-dry vacuum cleaner and the like. In the illustrated example, the base **102** of the surface cleaning apparatus **100** includes a surface cleaning head **104** and an elongated rigid wand **106**. In this configuration, the surface cleaning apparatus **100** can be used to clean a floor or other surface in a manner analogous to a conventional upright-style vacuum cleaner. In alternate configurations, the surface cleaning apparatus **100** can be used alone (without wand **106** or with an accessory cleaning tool), such as to clean above-floor surfaces, or with the wand **106**, such as to clean hard-to-reach and/or above-floor surfaces, for example.

[0074] Referring to FIGS. **2** to **6**, as exemplified, the hand vacuum **100** has a front end **108**, a rear end **110**, an upper end **112**, and a lower end **114**. The hand vacuum **100** may include a main body **116**, a handle **118** extending from the main body **116**, an air treatment chamber **120** connected to the main body **116** (e.g., fixed, removably, pivotably, or semi-pivotably connected), a dirty air inlet **122**, a clean air outlet **124**, an airflow path **126** extending between the inlet **122** and outlet **124**, and a suction motor **128** to generate vacuum suction through the airflow path **126**. The suction motor **128** may be positioned in a suction motor housing **130** within the main body **116**. Optionally, the hand vacuum **100** may include a pre-motor filter **132** and/or a post-motor filter **134**. The pre-motor filter **132** and post-motor filter **134** may respectively be positioned in the main body **116** within a

pre-motor filter housing **136** and a post-motor filter housing **138**, which may be integrally formed as part of the main body housing **118** or removably positioned therein.

[0075] In the illustrated example, the dirty air inlet **122** is provided at the front end **108** and, as exemplified, may be positioned proximate the upper end **112**. The hand vacuum **100** includes an air inlet conduit **140**, which may be a generally linear member having a conduit axis **142** that is oriented in a longitudinal forward/rearward direction. As exemplified, the conduit axis **142** is generally horizontal when the hand vacuum **100** is oriented with the upper end **108** above the lower end **110**.

[0076] Referring to FIG. **6**, the air inlet conduit **140** includes a sidewall **144** extending between axially opposed ends, and the dirty air inlet **122** is provided at one of the axially opposed ends and a dirty air outlet **146** is provided at the other of the axially opposed ends. As exemplified, the dirty air inlet **122** and the dirty air outlet **146** are ports at the front and rear ends of the air inlet conduit **140**. As exemplified, the dirty air inlet **122** and, optionally, the dirty air outlet **146** may be positioned forward of the air treatment chamber **120**. Optionally, the end of the air inlet conduit **140** having the dirty air inlet **122** can be used as a nozzle to directly clean a surface. In use, the dirty air outlet **146** introduces dirty air into the air treatment chamber **120**. It will be appreciated that the air inlet conduit **140**, dirty air inlet **122**, and/or dirty air outlet **146** may be provided in different locations than those exemplified.

[0077] As exemplified in FIG. **6**, the air treatment chamber **120** may be provided at the front end **108** of the hand vacuum **100**, such as forward of the suction motor **128** and the main body. The air treatment chamber **120** has a front end **148**, an opposed rear end **150**, a sidewall **152** extending between the front and rear ends **148**, **150**, a lower end **154**, and an upper end **156**. The air treatment chamber **120** may further have a central longitudinal axis **158** extending between the front and rear ends **148**, **150**. As used herein, “central” may mean the axis is centrally located with respect to the lateral direction of the hand vacuum **100** and/or centrally located with respect to the vertical direction when the hand vacuum **100** is oriented with the upper end **112** above the lower end **114**. Accordingly, the axis **158** may be radially located centrally inwardly of the sidewall **152**. The longitudinal axis **158** may be parallel to the conduit axis **142** of the air inlet conduit **140**. As exemplified in FIG. **6**, when the hand vacuum **100** is oriented with the upper end **112** above the lower end **114**, the longitudinal axis **158** of the air treatment chamber **120** and, in some embodiments, the conduit axis **142** of the air inlet conduit **140**, may be oriented horizontally.

[0078] As exemplified, dirty air enters the air treatment chamber **120** through a chamber air inlet **160** provided at the front end **148** and exits through a chamber air outlet **162** provided at the rear end **150**. The chamber air inlet **160** may be provided in a front end wall **166** of the air treatment chamber **120** and define an inlet port **164**, which may be fluidly connected to the dirty air outlet **146** of the air inlet conduit **140**. As exemplified, the inlet port **164** is also the outlet port of the air inlet conduit **140**.

[0079] The chamber air outlet **162** may be provided in a rear end wall **168** of the air treatment chamber **120** and may comprise a port in the rear end wall **168** of the air treatment chamber **120**. As exemplified, the chamber air outlet **162** also comprises a porous separator **170**, such as a screen or mesh, which may be positioned over the port of the chamber air outlet **162**. As exemplified, porous separator **170** optionally extends axially inwardly into the air treatment chamber **120** from the rear end wall **168**. The airflow path **126** may pass through the inlet port **164** into the air treatment chamber **120** and through the porous separator **170** to separate at least some particulate matter from the air as it travels to the chamber air outlet **162**.

[0080] The air treatment chamber **120** may be a non-cyclonic treatment chamber and may be a turbulent flow treatment chamber, which may include one or more elements for inducing turbulent flow in the airflow path **126** between the chamber air inlet **160** and the chamber air outlet **162**. Due to the changes in speed and/or direction of air undergoing turbulent flow, particulate matter will be disentranced from the air flow but will still move within the air treatment chamber **120**. Optionally,

some or all of air may be directed to flow over or past part or all of the porous separator **170** thereby removing particulate matter from the outer surface of the porous separator **170** and preventing or reducing the buildup of particulate matter on the porous separator **170**. By preventing or reducing the buildup of particulate matter on the porous separator **170**, the openings in the porous separator will not become clogged thereby enabling the volume of air passing through the air treatment chamber **120** to remain relatively constant during a cleaning operation.

[0081] As discussed subsequently herein, a non-cyclonic air treatment chamber **120** includes one or more of a non-return valve **172** provided at the chamber air inlet **160**, which directs air entering the air treatment chamber towards a sidewall (see e.g., FIG. 6), a baffle plate **174** positioned rearward of the chamber air inlet **160** (see e.g., FIG. 6), an air impervious cover **176** positioned at the rear end **150** of the air treatment chamber **120** (see e.g., FIGS. 20A to 20C) and an axially extending rib provided on an inner surface of the sidewall **152** of the air treatment chamber **120** (see e.g., FIGS. 19A and 19B). Each of the non-return valve **172**, the baffle plate **174**, the air impervious cover **176**, and the axially extending rib **178** may function to direct, deflect, divide, and/or disrupt the airflow through the air treatment chamber **120** to induce, e.g., turbulent flow therein.

[0082] Referring again to FIG. 6, as exemplified, the air treatment chamber **120** comprises a volume **180** defined by the sidewall **152** of the air treatment chamber **120**. As discussed in more detail subsequently, in use, the airflow in the volume **180** will result in particulate matter continually moving in the volume during a cleaning operation such that the particulate matter will settle onto a surface of the volume **180** only when the suction motor is deenergized. This differs to a cyclone chamber wherein particulate matter is entrained from the air flow and collected in a dirt collection chamber while cyclonic flow continues in the cyclone chamber. When a cleaning operation is terminated (i.e., the suction motor is deactivated), then the air flow through the air treatment chamber **120** will terminate and particulate matter will settle onto a surface defining the volume **180** of the air treatment chamber **120**.

[0083] At least a portion of the air treatment chamber **120** can be openable for emptying the volume **180**. For example, as exemplified in FIGS. 7 and 8, the sidewall **152** of the air treatment chamber **120** has an openable portion **182** (the lower portion as exemplified) and a stationary portion (the upper portion of the air treatment chamber **120** as exemplified). As exemplified, the openable portion **182** may be movably connected to the main body **116**, e.g., at the rear end **150** of the air treatment chamber **120**, such as by being pivotably connected, using any suitable mechanism including a hinge or other suitable device. The openable portion **182** may be movable between a closed position, in which the air treatment chamber **120** is closed (see e.g., FIGS. 2 and 3), and an open position, in which the air treatment chamber **120** is open and the volume **180** is emptiable (see e.g., FIGS. 7 and 8). It will be appreciated that any other opening design may be used. For example, as discussed subsequently, the air treatment chamber **120** may have a front and/or rear openable wall.

[0084] Optionally, as exemplified in FIGS. 7 and 8, the rear end wall **168** and the porous separator **170** may move with the openable portion **182** between the open and closed positions. The rear end wall **168** and the porous separator **170** moving with the openable portion **182** may advantageously allow enhanced access to the porous separator **170** to remove any buildup of particulate matter thereon.

[0085] Optionally, once opened, the openable portion together with the rear end wall **168** and the porous separator **170** may be removed (e.g., slid forwardly) to provide access to the pre-motor filter **132** in the pre-motor filter housing **136**, which may be located rearward of the air treatment chamber **120**, for removal and inspection, cleaning, or replacement of the pre-motor filter **132**.

[0086] Optionally, the openable portion **182** can be secured in the closed position using any suitable type of locking mechanism, including a latch mechanism that can be released by a user. The actuator for opening/releasing the openable portion **182** can be provided on the air treatment chamber **120** itself, on the main body **116**, or on any other portion of the hand vacuum **100**. This

may enable the hand vacuum **100** to be manually emptied by the user.

[0087] As another example, as exemplified in FIGS. **2** and **5**, the front end wall **166** of the air treatment chamber **120** may have an openable door **184** below the air inlet conduit **140**. The openable door **184** may be movably connected, such as pivotably connected, to the front end wall **166** of the air treatment chamber **120** using any suitable mechanism, including a hinge or other suitable device. The openable door **184** may be movable between a closed position, in which the air treatment chamber **120** is closed (see e.g., FIGS. **2**, **5**, and **9B**), and an open position, in which the air treatment chamber **120** is open and the volume **180** therein is emptiable (see e.g., FIG. **9C**).

[0088] Optionally, the openable door **184** can be secured in the closed position using any suitable type of locking mechanism, including a latch mechanism that can be released by a user (e.g., as described with respect to the openable portion **182**). Alternatively, the openable door **184** may be opened automatically. For example, FIG. **9A** shows the hand vacuum **100** docked at a docking station **186**. As exemplified, the docking station **186** includes a station base **188** and an upright conduit **190** extending upwardly from the station base **188**. The hand vacuum **100** is docked to a vacuum mount **192** of the docking station **186** at an upper end of the upright conduit **190**. The upright conduit **190** may extend to an elevation such that, as shown, the hand vacuum **100** may dock at the docking station **186** while attached to the wand **106** and surface cleaning head **104**. Once docked or as the hand vacuum **100** is docked, the openable door **184** may automatically move from the closed position (see e.g., FIG. **9B**) to the open position (see e.g., FIG. **9C**), placing the air treatment chamber **120** in fluid communication with the upright conduit **190**. Subsequently, a suction motor (not shown) in the station base **188** may generate suction to draw the contents of the air treatment chamber **120** through the opening of the openable door **184**, through the upright conduit **190**, and into a collection chamber (not shown) within the station base **188**. This may enable the hand vacuum **100** to be automatically emptied by the docking station **186**.

[0089] Referring again to FIG. **6**, after exiting the air treatment chamber **120** through the chamber air outlet **162**, the airflow path **126** may extend downstream to the suction motor **128** through any intervening elements, such as the pre-motor filter **132**. Optionally, the air treatment chamber **120** may be a first stage air treatment chamber, and the hand vacuum cleaner **100** may further include a second treatment stage (not shown) downstream of the first stage air treatment chamber (e.g., in series with each other such that air exiting the first stage air treatment chamber flows into the second stage air treatment chamber) and upstream of the suction motor **128**. In such embodiments, the second treatment stage may include a single cyclone chamber, a plurality of cyclones arranged in parallel, or may be non-cyclonic, such as a filter bag or a turbulent flow chamber as described with respect to the air treatment chamber **120**, for example.

[0090] The suction motor **128** used in the hand vacuum **100** may be of any suitable design and configuration that is sufficient to impart a desired air flow through the hand vacuum **100**. For example, the suction motor **128** may include a fan and/or impeller (i.e., a motor and fan assembly), which rotates about a motor axis **194** of rotation, to help generate the desired airflow. As exemplified, when the hand vacuum **100** is oriented with the upper end **112** above the lower end **114** in the orientation of FIG. **6**, the motor axis of rotation **194** of the suction motor **128** may be oriented horizontally. The motor axis **194** may extend in the same direction as the longitudinal axis **158** of the air treatment chamber **120** and/or the conduit axis **142** of the air inlet conduit **140** and, in some embodiments, one or both of the longitudinal axis **158** and the conduit axis **142** may be coaxial with the longitudinal axis **158**.

[0091] As exemplified in FIGS. **2** to **4**, the clean air outlet **124** includes a grill **196**, which may be oriented such that air exiting the clean air outlet **124** travels generally upwardly, generally laterally outwardly, and/or at an inclined angle generally rearwardly from the hand vacuum **100**. Such directional airflow exiting the clean air outlet **124** may beneficially avoid the exhaust airflow impacting the user or the surface to be cleaned. The clean air outlet **124** may be at any position rearward of the dirty air inlet **122** such as proximate the rear end **110**, as shown, or at any other

position intermediate the front and rear ends **108**, **110**. The clean air outlet **124** may further be at any position between the upper and lower ends **112**, **114**, such as proximate the upper end **112** as shown. Any other position of the clean air outlet **124** may be possible, such as in or below handle **118**.

[0092] Optionally, one or more pre-motor filters **132** may be provided in the hand vacuum **100** at any position in the airflow path **126** upstream from the suction motor **128**, such as in the pre-motor filter housing **136** in the main body **116**. The pre-motor filter **132** may be formed from foam or any other suitable physical, porous filter media and have any suitable shape, such as a donut filter with a filter media defining a sidewall and an open internal volume defining a downstream filter side wherein the filter may be cylindrical, conical, frusto-conical, or a generally flat, slab-like filter. An optional felt filter layer can be provided on one side of the pre-motor filter **132** (i.e., on the upstream side or, preferably, adjacent the downstream side).

[0093] Optionally, one or more post-motor filters **134** may, alternatively or in addition to the pre-motor filter **132**, be provided in the hand vacuum **100** at any position in the airflow path **126** downstream from the suction motor **128**, such as in the post-motor filter housing **138** in the main body **116** or in the handle **118** (e.g., if the clean air outlet **124** is provided in the handle **118**). The post-motor filter **134** may be a physical foam media filter or may be any other suitable physical porous filter media, including, for example, a felt filter, a HEPA filter, a paper filter, other physical filter media, an electrostatic filter, and the like.

[0094] The hand vacuum **100** in accordance with the example embodiments described herein may include any suitable type of carry handle as part of the main body **116**. As exemplified in FIGS. 2 to 6, the handle **118** may be located proximate the rear end **110** of the hand vacuum **100**. The handle **118** may be a pistol-grip style handle having an elongate hand grip portion **198** extending downwardly and rearwardly along a hand grip axis **200** (see e.g., FIG. 6) between an upper end **202**, which is coupled to the sidewall of the main body **116**, and a lower end **204**. When the hand vacuum **100** is oriented so that the upper end **112** is disposed above the lower end **114**, the hand grip axis **200** may form an acute angle, relative to the vertical direction. The angle can be any suitable angle, and preferably is between about 5-85°, and may be between about 10-40°.

[0095] As exemplified, the lower end **204** of the handle **118** may include a lower member **206**, which may function as a stand and/or housing for various elements of the hand vacuum **100**. For example, power may be supplied to the hand vacuum **100** by one or more onboard energy storage members **208** (see e.g., FIG. 6), which may be housed, for example, in the lower member **206**. Additionally, or alternatively, the energy storage members **208** may be housed in the hand grip portion **198** of the handle **118** and/or in the main body **116**. Positioning energy storage members **208** at two or more locations may help distribute the weight of the energy storage members **208** and may affect the hand feel and/or perceived balance of the hand vacuum **100**.

[0096] Referring to FIG. 1, energy storage members **208** may optionally also be provided within the wand **106**, the surface cleaning head **104**, and/or other auxiliary tools. Providing energy storage members **208** in the wand **106** and/or the surface cleaning head **104** may help provide some additional power when the hand vacuum **100** is connected to the wand **106** and/or to the surface cleaning head **104** via the wand **106**. For example, the extra power in the additional energy storage members **208** may be used to help power a brush motor, lights, or other such features that require power in the surface cleaning head **104** and/or help power the suction motor **128**. If the energy storage members **208** housed in or on the wand **106** and/or the surface cleaning head **104**, the additional mass of those energy storage members **208** can be left behind when a user detaches the hand vacuum **100** for above floor cleaning. This may help reduce the weight of the hand vacuum **100**, while still providing a desired level of power when operating in the floor cleaning mode. Optionally, the energy storage members **208** housed in or on the wand **106** and/or the surface cleaning head **104**, together with the energy storage members provided in the hand vacuum **100** can provide at least some power to the suction motor **128** when the hand vacuum **100** is mounted to the

wand **106** and one or both may optionally at least some power to the surface cleaning head **104**. This may help provide longer run times, higher suction levels or both as compared to only using the power supplied from the hand vacuum **100**.

[0097] The energy storage members **208** may include, for example, one or more batteries and/or battery packs, which may be any type including, for example, solid state batteries, or one or more capacitors, super capacitors or ultra capacitors. If the energy storage member(s) **208** is/are a battery pack, each battery pack may include any suitable number of cells, and may include, for example, 3 cell 18560 lithium ion batteries. If two battery packs are connected in series, they may create a 6 cell 22V Li-ion power source. Any number of cells may be used to create a power source having a desired voltage and current, and any type of battery may be used, including NiMH, alkaline and the like.

[0098] Optionally, the batteries and battery packs may be rechargeable or may be replaceable, non-rechargeable batteries. For example, if the batteries and battery packs are rechargeable, the docking station **186** exemplified in FIG. **9A** may be configured to charge the energy storage members **208** when the hand vacuum **100** is docked to the vacuum mount **192**. As another example, the hand vacuum **100** may optionally include an electrical cord (not shown) that can be connected to a standard wall electrical outlet. In such examples, power for the hand vacuum **100** may be supplied via the electrical cord from a wall socket, and the suction motor **128** may run on AC power supplied from the wall socket. Optionally, the electrical cord may charge the energy storage members **208** and be detachable from the hand vacuum **100**, allowing for both corded and cordless operation of the hand vacuum **100**. Optionally, in alternate embodiments, the energy storage members **208** may be omitted and the power source for the hand vacuum **100** may be solely provided via the electrical cord.

[0099] In addition to functioning as a nozzle, the air inlet conduit **140** can be connected to the downstream end of any suitable accessory tool such as a rigid air flow conduit (e.g., the wand **106**), crevice tool, mini brush or the like. The hand vacuum **100** can include an electrical connector **210** (see e.g., FIG. **2**), which may be provided proximate the front end **108**, such as on the air inlet conduit **140**. The downstream end of the accessory, such as the wand **106**, can include another electrical connector (not shown), which may be detachably matingly connectable to the electrical connector **210** of the hand vacuum **100**. The electrical connectors **210** may be of any suitable configuration, such as mating pins and sockets. Power can thereby be communicated between the surface cleaning head **104** and the hand vacuum **100** via the wand **106**.

[0100] For example, FIG. **1** shows an exemplary embodiment of a stickvac cleaning apparatus (e.g., a stickvac) that incorporates hand vacuum **100** and is configured so that the air inlet conduit **140** is directly mechanically, and optionally electrically, connected to an upper end **212** of the wand **106**. A lower end **214** of the wand **106** is pivotally, and optionally pivotally and steeringly, connected to the surface cleaning head **104**. In this arrangement, the handle **118** can be used to manipulate the hand vacuum **100** when detached from the wand **106**, and can be used to manipulate the combination of the hand vacuum **100** and the wand **106**, or the combination of the hand vacuum **100**, wand **106**, and surface cleaning head **104** (i.e. the stickvac), depending on the mode or configuration in which the hand vacuum **100** is used.

[0101] The wand **106** may be any suitable member that can provide the desired structural connection and airflow communication between the hand vacuum **100** and the surface cleaning head **104**. Preferably, the wand **106** is a rigid air flow conduit that provides the fluid communication between the surface cleaning head **104** and the hand vacuum **100**. The wand **106** may be made from metal, plastic and/or any other suitable material.

[0102] In the illustrated example, the surface cleaning head **104** includes a body **216**, a pair of rear wheels (not shown) connected to the body **216** to rollingly support the surface cleaning head **104** above a surface to be cleaned, and a cleaning head dirty air inlet **218** in the downward facing lower body surface. The surface cleaning head **104** also includes an upflow duct **220** that is pivotally

connected to the body **216**. The lower end **214** of the wand **106** can be connected to the upflow duct **220**, whereby the wand **106** is movable relative to the surface cleaning head **104**. In the illustrated example, the upflow duct **220** is in fluid communication with the cleaning head dirty air inlet **218**. The upflow duct **220** is configured to receive the lower end **214** of the wand **106** (or optionally vice versa) and to fluidly connect the surface cleaning head **104** to the hand vacuum **100**. Optionally, the lower end **214** of the wand **106** can be detachably connected to the upflow duct **220**, using any means, including a friction fit, suitable latch, locking mechanism or the like. Providing a detachable connection may allow the wand **106** to be separated from the surface cleaning head **104** for maintenance and/or for use in above floor cleaning. In such a configuration, the lower end **214** of the wand **106** can function as an auxiliary dirty air inlet, and the wand **106** as an extension of the air inlet conduit **140**. It will be appreciated that any surface cleaning head may be used.

[0103] Preferably, as previously discussed, the hand vacuum **100** can be detachably connected to the opposing upper end **212** of the wand **106**, for example using latch, so that the hand vacuum **100** can be detached and used independently from the wand **106** and/or surface cleaning head **104**.

Providing detachable connections at both ends of the wand **106** may help facilitate use of the hand vacuum **100** in at least three different operating modes:

(i) an upright cleaning mode in which both the surface cleaning head and hand vacuum are attached to the wand and there is an airflow path extending from the dirty air inlet in the surface cleaning head to the hand vacuum and including the wand; (ii) a first above floor cleaning mode in which the wand is detached from the surface cleaning head and an airflow path extends from the auxiliary dirty air inlet of the wand to the hand vacuum cleaner; and

(iii) a second above floor cleaning mode in which the hand vacuum is detached from the upper end of the wand and the nozzle is used to directly clean a surface and/or is connected to one or more auxiliary cleaning tools (such as a hose, crevice tool, upholstery brush and the like).

[0104] Optionally, the hand vacuum **100** may include one or more information display devices, such as one or more lights, to provide information to a user. For example, as shown in FIG. **1**, the information display device of the hand vacuum **100** is a display screen **222**, which may be an LCD screen, an LED screen, an OLED screen and the like. While the display screen **222** is provided on the rear end **110** of the hand vacuum **100**, it will be appreciated that other the display screen **222** may be provided on the upper end **112** of the hand vacuum **100** or any other suitable location allowing the user to monitor the information displayed on the screen **222**. The information displayed on the screen **222** or other suitable information display device(s) may indicate status information of the hand vacuum **100** (e.g., power mode, operation mode), the suction motor **128** (e.g., on/off, current power level, overheating), the energy storage members **208** (e.g., charge level) and/or other electrical elements of the hand vacuum **100**.

[0105] Optionally, the information display device may also provide information related to an accessory tool of the like that is connected to the hand vacuum **100**, such as a tool or the wand **106** and/or surface cleaning head **104**. The information displayed may be customized for each type of accessory tool or apparatus that can be connected to the hand vacuum **100**. For example, the screen **222** of FIG. **1** can optionally be configured to show information about the accessory tool or apparatus is connected to the hand vacuum **100** (e.g., via the electrical connector **210**), so that the same screen can be used for multiple apparatuses. This may reduce the need to provide screens or the like on each separate apparatus that can be connected to the handle.

[0106] Optionally, as exemplified in FIG. **1**, a power switch **224** that controls operation of the suction motor **128** (on/off, variable power levels, or both) can be provided on the rear end **110** of the hand vacuum **100**, for example by establishing a power connection between the energy storage members **208** and the suction motor **128** (or other portion of the hand vacuum). The power switch **224** can be provided in any suitable configuration, including as a button (as shown), a rotary switch, or a sliding switch, for example. The display screen **222** may also be a touch screen, and the power switch **224** can be provided as a touch button in the display screen **222**. The power switch

224 may be provided at any other suitable location, such as on or near the handle **118** (e.g., a trigger-type actuator on a forward side at upper end **202** of handle **118**) such that it can be actuated by a finger (e.g., thumb or index finger) of the user on the same hand as is gripping the hand grip portion **198** of the handle **118**, allowing true one-handed operation of the hand vacuum **100**.

[0107] The power switch **224** may also be configured to control other aspects of the hand vacuum **100** (e.g., brush motor on/off, etc.) or may be configured entirely as a control switch that controls some functions of the hand vacuum **100** but does not control the suction motor **128**. Optionally, the power switch **224**, or an additional designated power switch, may be used to provide different functions for different powered devices (e.g., the surface cleaning head **104**), if the hand vacuum **100** is connected to different devices. For example, the power switch **224** may be used to select different power profiles (high power vs. low power), or other device functions. Optionally, the function of the power switch **224** may be displayed on a suitable display device, such as the display screen **222** described herein. For example, the power switch **224** may be operable to control both the suction motor **128** and a brush motor in the surface cleaning head **104**, along with lights and any other electrified elements provided on the wand **106**, surface cleaning head **104**, or any other tool or attachment.

[0108] Optionally, the hand vacuum cleaner **100** may be configurable in two or more different operating modes, having different power profiles. For example, the suction motor **128** in the hand vacuum **100** may be operable at a low power mode and a high power mode, each providing different levels of suction and air flow through the hand vacuum **100**. Switching between such power modes may be done manually by a user, in some embodiments, or may be done automatically based on the configuration or operation of the hand vacuum in other embodiments. For example, the apparatus may be operable to automatically change power modes based on the detected presence and/or amount of dirt on the surface to be cleaned. The hand vacuum **100** may also include a manual option for a user to override the automatic changes.

Porous Separator Positioned at the Chamber Air Outlet

[0109] A hand vacuum cleaner using any one or more aspects discussed herein, including one or more of the other aspects set out herein including one or more of the non-return valve flap that closes an inlet port of the chamber air inlet, the baffle plate positioned in the air treatment chamber rearward of the chamber air inlet, the air impervious cover positioned at the rear end of the air treatment chamber and overlying the porous separator, the axially extending rib provided on an inner surface of the sidewall of the air treatment chamber, and the bleed air inlet port provided through the sidewall of the air treatment chamber, may have the porous separator positioned at the chamber air outlet.

[0110] As described previously, the porous separator **170** may be provided as part of the chamber air outlet **162** and may separate at least some particulate matter from the airflow as the airflow passes through the porous separator **170** while travelling to the outlet port of the chamber air outlet **162**. The porous separator **170** may be any shape, such as frusto-conical (see e.g., FIG. **6**), domed or semi-spherical (see e.g., FIG. **10A**), generally planar (see e.g., FIG. **11A**), or conical (see e.g., FIG. **12A**), for example. As shown in FIGS. **6** and **10A** to **10C**, the porous separator **170** may include a rigid screen **226** through which the airflow passes. The rigid screen **226** may have a plurality of pores, and the pores may have any size suitable for permitting the passage of air therethrough while restricting the passage of particulate matter. The rigid screen **226** may be self-supporting or, as shown, supported by a plurality of frame members **227** which extend from a base **228**.

[0111] Optionally, as exemplified, the porous separator **170** may include a ring **229** to which the base **228** may be removably connected (e.g., threadedly, snap-fit, etc.). As exemplified, the ring **229** has an axially or generally axially extending sidewall that is air impervious. The ring **229** may be connected to the rear end wall **168** of the air treatment chamber **120** over the chamber air outlet **162**. In such examples, the screened portion of the porous separator **170** (i.e., the rigid screen **226**,

frame members **227**, and base **228**) may advantageously be removed from the ring **229** for inspection, cleaning, and/or replacement. Further, porous separators **170** of various shapes may advantageously be interchangeably connected to the ring **229** to change the airflow dynamics and/or collection capacity of the air treatment chamber **120**. For example, porous separators **170** having differently sized openings may be interchangeable connected to the ring **229**. In alternate embodiments, the ring **229** may be omitted and the base **228** may be directly removably connected to the rear end wall **168** or the base **228** may be integrally formed with the ring **229**, which may itself be directly removably connected to the rear end wall **168**.

[0112] In some embodiments, such as exemplified in FIGS. **6**, **10B** to **10C**, and **12B** to **12C**, the porous separator **170** may extend forwardly from the rear end wall **168** into the air treatment chamber **120**. Optionally, the porous separator **170** may extend forwardly from the rear end wall **168** along the longitudinal axis **158** of the air treatment chamber **120**. In some embodiments, such as exemplified in FIGS. **11B** to **11C**, if the porous separator **170** is generally planar, it may extend in a plane that is generally perpendicular to the longitudinal axis **158** of the air treatment chamber **120**.

[0113] In some embodiments, the ring **229** (or base **228**, if the ring **229** is omitted) of the porous separator **170** may have a smaller outer diameter than the rear end wall **168**. In such embodiments, the porous separator **170** is spaced radially inwardly from the sidewall **152** of the air treatment chamber **120**. In alternate embodiments, the base **228** or ring **229** of the porous separator **170** may have substantially the same outer diameter as the rear end wall **168** such that the rear end wall **168** is substantially covered by the porous separator **170**. This may depend, for example, on the size of the chamber air outlet **162** and/or the desired size of the porous separator **170**, such as for the desired amount of filtering surface area of the rigid screen **226**.

Non-Return Valve Flap that Closes an Inlet Port of the Chamber Air Inlet

[0114] A hand vacuum cleaner using any one or more aspects discussed herein, including one or more of the other aspects set out herein including one or more of the porous separator positioned at the chamber air outlet, the baffle plate positioned in the air treatment chamber rearward of the chamber air inlet, the air impervious cover positioned at the rear end of the air treatment chamber and overlying the porous separator, the axially extending rib provided on an inner surface of the sidewall of the air treatment chamber, and the bleed air inlet port provided through the sidewall of the air treatment chamber, may have the non-return valve flap that closes an inlet port of the chamber air inlet.

[0115] The non-return valve **172** in the hand vacuum **100** as described herein may close the inlet port **164** of the air treatment chamber **120** when the hand vacuum **100** is turned off. This may advantageously prevent particulate matter collected within the volume **180** from escaping the air treatment chamber **120** through the inlet port **164** and falling from the hand vacuum **100** through the air inlet conduit **140** (and wand **106**, if attached), such as when the hand vacuum **100** is oriented with the rear end **110** above the front end **108**.

[0116] In accordance with this aspect and as exemplified in FIG. **15**, the non-return valve **172** is configured or operable to direct (deflect) incoming dirty air exiting the inlet port **164** to flow generally linearly in the direction of the longitudinal axis towards the porous separator **170**, to then reverse flow and travel past the porous separator **170** so as to wash the porous separator **170** (e.g., as the air travels forwardly). Subsequently, the air again reverses direction and creates a flow, which may be turbulent, towards and then through the porous separator **170**. Accordingly, the opening created by a non-return valve flap **230** of the non-return valve **172** (i.e., the opening between the inlet port **164** and the side of the non-return valve flap **230** which faces the inlet port **164** in the closed position) directs the incoming air towards a desired portion of the sidewall **152** of the air treatment chamber **120** (see for example the airflow indicated in FIG. **13B**. As exemplified, and in the orientation of FIG. **13B**, the incoming air is directed downwardly to a portion of the sidewall **152** that is rearward of the inlet port **164**. The air is then redirected by the wall to travel

generally rearwardly towards the porous separator **170**. Accordingly, it will be appreciated that the airflow in the volume **180** will result in particulate matter continually moving in the volume during a cleaning operation and that the particulate matter will settle onto a surface of the volume **180** only when the suction motor is deenergized.

[0117] It will be appreciated that, in the embodiment of FIG. **6**, if the non-return valve flap **230** is thin and planer as exemplified and if the non-return valve flap **230** were to rotate open 90° or more, then air exiting the air inlet conduit **140** would travel generally linearly (rearwardly) to the porous separator **170**. In such a case, the air flow would directly impinge upon the porous separator **170** resulting in some particulate matter being lodged in the opening of the porous separator **170** and additional particulate matter collecting on the outer surface of the porous separator **170**.

[0118] Referring now to FIGS. **13A** to **14**, the non-return valve **172** of the hand vacuum **100** includes a non-return valve flap **230** having a first end **232** (shown as the upper end) and an opposed end **234** (shown as the lower end). The first end **232** may be mounted to the air treatment chamber **120** at any location suitable for positioning the non-return valve flap **230** such that it may close the inlet port **164** of the chamber air inlet **160** when the suction motor is deenergized. For example, the first end **232** (e.g., the upper end in the orientation of FIG. **13B**) may be movably mounted, such as pivotably or rotatably, proximate the inlet port **164**. Optionally, if the inlet conduit **140** is provided adjacent the sidewall **152** (e.g., at the upper end of the volume **180** as exemplified in FIG. **6**, then the non-return valve flap **230** may be moveably mounted at a location adjacent a portion of an inner surface of the sidewall **152**. In such examples, the first end **232** may optionally be mounted to the inner surface of the sidewall **152** of the air treatment chamber **120** (e.g., at the upper end **156**) or to the front end wall **166** of the air treatment chamber **120** (e.g., above the inlet port **164**) proximate the dirty air outlet **146**.

[0119] As described subsequently, the mounting location of the first end **232** may depend, for example, on the desired direction of the non-return valve flap **230** to direct/deflect the airflow entering the air treatment chamber **120**. In the illustrated example, the first end **232** is pivotably mounted to the air treatment chamber **120** adjacent the inner surface of the sidewall **152** at the upper end **156** of the air treatment chamber **120**.

[0120] It will be appreciated that, in alternate embodiments, the non-return valve flap **230** may be mounted at the lower end of the volume **180** (in the orientation of FIG. **6**) or the inlet conduit **140** may be centrally located such that conduit axis **142** is adjacent or coaxial with the longitudinal axis **158**.

[0121] As exemplified, the non-return valve flap **230** may be moveable from a closed position (see e.g., FIG. **13A**), in which it closes the inlet port **164**, to an open position (see e.g., FIG. **13B**), in which the opposed end **234** of the non-return valve flap **230** rotates inwardly and towards the sidewall **152** (i.e., upwardly, in the illustrated example) whereby the inlet port **164** is opened.

[0122] In any embodiment, the non-return valve flap **230** may be biased to the closed position by a biasing member such as, for example, a torsion spring or any other suitable biasing means. For example, if the non-return valve flap **230** is made of a resilient material, the non-return valve flap **230** may bias itself to the closed position. In use, when the hand vacuum **100** is switched on, the suction force generated by the suction motor **128** may overcome the force of gravity and/or the biasing force of the biasing member and cause the non-return valve flap **230** to move from the closed position to the open position. Conversely, when the suction motor **128** is turned off such that suction force is removed, the non-return valve flap **230** may return to the closed position under the force of gravity or the biasing force from a biasing member.

[0123] The non-return valve flap **230** may be dimensioned such that, in the closed position, it fully covers the inlet port **164**. For example, the non-return valve flap **230** may have a greater diameter than that of the inlet port **164**.

[0124] Optionally, a forward facing surface of the non-return valve flap **230** may include a resiliently compressible material which, under force of the biasing member, contacts and

compresses against the front end wall **166** of the air treatment chamber **120** around the inlet port **164** and/or the sidewall **144** at the end of the air inlet conduit **140** having the dirty air outlet **146**. In this way, the resiliently compressible material may form an improved seal, preventing escape of any collected particulate matter from the air treatment chamber **120** when the non-return valve flap **230** is in the closed position. Alternatively, the resiliently compressible material may be provided on the front end wall **166** around the inlet port **164** and/or sidewall **144** at the end of the air inlet conduit **140** having the dirty air outlet **146**.

[0125] The open position of the non-return valve flap **230** may be any rotational angle from the closed position. For example, as exemplified in FIG. **13B**, the non-return valve flap **230** may rotate open less than 90° (e.g., less than 90°, 80°, 70°, 60°, 50° or less than 45°) to the open position. In any embodiment, the rotational angle may be regulated by a physical stop, such as a physical stop extending inwardly from the inner surface of the sidewall **152** of the air treatment chamber **120** (e.g., from the upper end, in the illustrated example). In such embodiments, the physical stop may contact a rearward facing surface of the non-return valve flap **230** and prevent any further rotation. In alternate embodiments, the rotational angle may be regulated by the biasing member. In such embodiments, the suction force generated by the suction motor **128** may be sufficient to overcome the biasing force of the biasing member to move the non-return valve flap **230** from the closed position, and the suction force and biasing force may reach an equilibrium at a predetermined rotational angle defining the open position.

[0126] It will also be appreciated that the forward facing surface of the non-return valve flap **230** need not be planar as exemplified. For example, the forward facing surface of the non-return valve flap **230** may be shaped to direct air downwardly (in the orientation of FIG. **6**) even if the non-return valve flap **230** opens 90°. Accordingly, the extent to which the non-return valve flap **230** opens may be determined based on the desired location on the sidewall **152** that the air is to be directed to and the shape of the forward facing surface of the non-return valve flap **230**.

[0127] In the open position, air exiting the inlet port **164** is directed in a non-tangential manner across the volume of the chamber towards a portion of the sidewall **152** that is opposed to the location of the first end **232** of the non-return valve flap **230** (e.g., the portion adjacent which the first end **232** of the non-return valve flap **230** is mounted) in a direction that is generally transverse to the longitudinal axis **158**. Referring to FIG. **13B**, in the example shown, air exiting the inlet port **164** is directed towards the inner surface of the sidewall **152** at the lower end **154** of the air treatment chamber **120**. It will be appreciated that the mounting location of the first end **232**, and thus the rotational direction of the second end **234** during opening, may dictate the direction that the non-return valve flap **230** directs/deflects the airflow entering the air treatment chamber **120**.

Baffle Plate Positioned in the Air Treatment Chamber Rearward of the Chamber Air Inlet

[0128] A hand vacuum cleaner using any one or more aspects discussed herein, including one or more of the other aspects set out herein including one or more of the porous separator positioned at the chamber air outlet, the non-return valve flap that closes an inlet port of the chamber air inlet, the air impervious cover positioned at the rear end of the air treatment chamber and overlying the porous separator, the axially extending rib provided on an inner surface of the sidewall of the air treatment chamber, and the bleed air inlet port provided through the sidewall of the air treatment chamber, may have the baffle plate positioned in the air treatment chamber rearward of the chamber air inlet.

[0129] Similar to the non-return valve **172**, a baffle plate **174** provided in the hand vacuum **100** as described herein may advantageously block incoming dirty air exiting the inlet port **164** from travelling directly from the chamber air inlet **160** to the chamber air outlet **162**, which may otherwise result in particulate matter clogging or building up on the porous separator **170**.

[0130] If a non-return valve **172** is provided, then the baffle plate may be provided as a backup or to supplement the non-return valve **172** in preventing or inhibiting incoming dirty air exiting the inlet port **164** from travelling directly from the chamber air inlet **160** to the chamber air outlet **162**.

Alternately if a non-return valve **172** is not provided, then the baffle plate may induce a similar airflow as discussed with respect to the non-return valve **170** and as exemplified in FIG. **15**.

[0131] The baffle plate **174** is positioned in the air treatment chamber **120** rearward of the chamber air inlet **160**. In the position shown, the baffle plate **174** is spaced from and overlies the inlet port **164** of the air treatment chamber **120** such that the conduit axis **142** of the air inlet conduit **140** intersects the baffle plate **174**. The spacing of the baffle plate **174** from the inlet port **164** may be at least enough for the non-return valve flap **230**, if provided, to rotate from the closed position to the open position. In some embodiments, the baffle plate **174** may function as the physical stop defining the open position for the non-return valve flap **230**. In such embodiments, the baffle plate **174** may be spaced from the inlet port **164** such that the non-return valve flap **230** may rotate to the desired rotational angle defining the open position, at which angle the second end **234** of the non-return valve flap **230** may contact the front side **236** of the baffle plate **174**, preventing further rotation.

[0132] As exemplified in FIG. **6**, the hand vacuum **100** includes the baffle plate **174** having a front side **236**, a rear side **238**, a radial outer edge **240**, and a radial inner free edge **242**.

[0133] The baffle plate **174** is shaped and positioned such that the baffle plate **174** overlies the inlet port **164**. As exemplified, the radial outer edge **240** of the baffle plate **174** is provided on the inner surface of the sidewall **152** such that the baffle plate **174** follows the shape of the air treatment chamber **120**. The baffle plate **174** may extend from the sidewall **152** in a direction generally transverse to the longitudinal axis **158** of the air treatment chamber **120** to the free edge **242** such that, as exemplified, the baffle plate **174** may be generally semi-circular in a plane transverse to the longitudinal axis **158**. In this embodiment, the baffle plate **174** has a transverse length sufficient for the baffle plate **174** to overlie the inlet port **164** and the free edge **242** to be spaced from the lower side of the inlet port **164** (in the orientation of FIG. **6**). It will be appreciated that any other shape may be possible. For example, the radial outer edge **240** need not be at the inner surface of the sidewall **152** but may be located partially or fully radially inward thereof.

[0134] The baffle plate **174** may be sized such that the free edge **242** may be positioned at any elevation within air treatment chamber **120**. The transverse or vertical positioning of the free edge **242** may influence the degree of directional change in the airflow path **126** as air diverts around the baffle plate **174** while travelling from the chamber air inlet **160** to the chamber air outlet **162**. For example, when the hand vacuum **100** is oriented with the upper end **112** above the lower end **114**, the free edge **242** may be below or aligned with a lower end of the chamber air inlet **160** and may also be below or aligned with an upper end of the chamber air outlet **162**. Optionally, the free edge **242** may further be below or aligned with a lower end of the chamber air outlet **162**. In the illustrated example, the free edge **242** is below the lower end of the chamber air inlet **160** and below the upper end of the chamber air outlet **162**.

[0135] Referring now to FIGS. **16A** to **16C**, shown therein are various alternate baffle plates **174**. As shown, the free edge **242** of the baffle plate **174** has a first lateral end **244a** and a second lateral end **244b** and the free edge **242** is unattached to any portion of the sidewall **152** between the first and second lateral ends **244a**, **244b**. The free edge **242** may have any shape between the first and second ends **244a**, **244b**. For example, the free edge **242** may be linear such as shown in FIG. **16A**. In any embodiment, the free edge **242** may be shaped to further influence the airflow path **126** as it diverts around the baffle plate **174**.

[0136] For example, the free edge **242** may have one or more recesses **246** that extend radially outwardly (i.e., toward the upper end **156** of the air treatment chamber **120**), such as two spaced apart recesses **246** as exemplified in FIG. **16B**. As another example, the free edge **242** may have one or more protrusions that extend radially inwardly (i.e., toward the lower end **154** of the air treatment chamber **120**), such as one protrusion **248** as exemplified in

[0137] FIG. **16C**. The inclusion of one or more recesses or protrusions may split the airflow path **126** into two or more airflow paths **126** as it passes through the recess(es) or around the

protrusion(s). In the examples shown in FIGS. **16B** and **16C**, the recesses **246** and the protrusion **248** may split the airflow path **126** into two airflow paths **126** as it passes through the recesses **246** or around the protrusion **248**, such that the airflow path **126** may continue as exemplified and previously described with respect to the non-return valve flap **230** with reference to FIG. **15**.

[0138] Alternatively, or in addition to the inclusions of recesses **246** and/or protrusions **248** provided on the free edge **242** of the baffle plate **174**, the front side **236** of the baffle plate **174** may include one or more forwardly extending portions to split the airflow as it impacts on the front side **236**. For example, as exemplified in FIGS. **17A** and **17B**, the front side **236** of the baffle plate **174** has one forwardly extending portion **250** that is centrally located between the first and second laterally spaced apart sides **244a**, **244b** of the free edge **242** and that extends axially forwardly from the front side **236** toward the inlet port **164**. In the illustrated example, the forwardly extending portion **250** has a triangular cross-section in a plane parallel to the longitudinal axis **158** of the air treatment chamber **120**. Any other cross-sectional shape suitable for splitting an incoming airflow may be possible. As exemplified, a tip of the forwardly extending portion **250** is centrally positioned proximate the inlet port **164** such that the conduit axis **142** of the air inlet conduit **140** extends through the tip. In the position shown, air entering the air treatment chamber **120** may be divided into two airflow paths **126** over the forwardly extending portion **250** before contacting the front side **236** of the baffle plate **174**.

[0139] It will be appreciated that, in other embodiments, the forwardly extending portion **250** may alternately or in addition be shaped to direct airflow downwardly (in the orientation of FIG. **6**) such that air exiting the inlet port **164** is directed to the sidewall as exemplified in FIG. **15**.

[0140] In embodiments wherein sidewall **152** of the air treatment chamber **120** includes the openable portion **182**, the free edge **242** of the baffle plate **174** may be positioned at an elevation above (e.g., adjacent) an upper end of the openable portion **182**. Similarly, in embodiments wherein front end wall **166** of the air treatment chamber **120** includes the openable door **184**, the free edge **242** of the baffle plate **174** may be positioned at an elevation above or aligned with an upper end of the openable door **184**.

[0141] In the examples shown in FIGS. **18A** to **18C**, the free edge **242** of each baffle plate **174** is above the upper end of both the openable portion **182** and the openable door **184**. Accordingly, as shown, the baffle plates **174** are sized to retain particulate matter rearward thereof while providing an unobstructed path for emptying the volume **180** of the air treatment chamber **120**.

[0142] Optionally, the baffle plate **174** may further be shaped to assist in emptying the volume **180** of the air treatment chamber **120**. For example, as shown in FIGS. **18B** and **18C**, the rear side **238** of the baffle plate **174** may be curved (see e.g., FIG. **18B**) or linearly sloped (see e.g., FIG. **18C**) such that the radial outer edge **240** on the rear side **238** is located further axially inwardly than the free edge **242** on the rear side **238**. In this way, the baffle plate **174** is shaped such that, when the dirty air inlet **122** faces downwardly, the rear side **238** of the baffle plate **174** directs dirt retained rearward of the baffle plate **174** towards the openable portion **182** (and/or openable door **184**, where provided).

Axially Extending Rib Provided on An Inner Surface of the Sidewall of the Air Treatment Chamber
[0143] A hand vacuum cleaner using any one or more aspects discussed herein, including one or more of the other aspects set out herein including one or more of the porous separator positioned at the chamber air outlet, the non-return valve flap that closes an inlet port of the chamber air inlet, the baffle plate positioned in the air treatment chamber rearward of the chamber air inlet, the air impervious cover positioned at the rear end of the air treatment chamber and overlying the porous separator, and the bleed air inlet port provided through the sidewall of the air treatment chamber, may have the axially extending rib provided on an inner surface of the sidewall of the air treatment chamber.

[0144] Providing an axially extending rib on the inner surface of the sidewall **152** of the air treatment chamber **120** may provide several advantages. For example, if a non-return valve flap

230 is provided to direct the incoming airflow towards an opposed portion of the inner surface of the sidewall **152** having the axially extending rib, the rib may assist in splitting the airflow into two airflow paths **126** as it contacts the rib and deflects off of the sidewall **152**, improving the production of the linear rearward airflow through the air treatment chamber **120**.

[0145] In the example shown in FIG. **19A**, the sidewall **152** of the air treatment chamber **120** has an axially extending rib **178** provided on the inner surface of thereof and extending between the front end **148** and the rear end **150** of the air treatment chamber **120**. If the air treatment chamber **120** includes the baffle plate **174**, then as exemplified in FIG. **19B**, the axially extending rib **178** may extend along the inner surface of the sidewall **152** from a location rearward of the baffle plate **174** to the rear end **150** of the air treatment chamber **120**.

[0146] While only one axially extending rib **178** is provided at the lower end **154** of the air treatment chamber **120** as shown in the illustrated examples, it will be appreciated that any number of axially extending ribs **178** may be provided at any location along the inner surface of the sidewall **152**.

[0147] It will be appreciated that any rib **178** may extend continually along part or all of the sidewall or it may be discontinuous (e.g., axially extending gaps may be provided between axially extending rib segments).

Air Impervious Cover Positioned at the Rear End of the Air Treatment Chamber and Overlying the Porous separator

[0148] A hand vacuum cleaner using any one or more aspects discussed herein, including one or more of the other aspects set out herein including one or more of the porous separator positioned at the chamber air outlet, the non-return valve flap that closes an inlet port of the chamber air inlet, the baffle plate positioned in the air treatment chamber rearward of the chamber air inlet, the axially extending rib provided on an inner surface of the sidewall of the air treatment chamber, and the bleed air inlet port provided through the sidewall of the air treatment chamber, may have the air impervious cover positioned at the rear end of the air treatment chamber and overlying the porous separator.

[0149] An air impervious cover **176** provided in the hand vacuum **100** as described herein is positioned and/or shaped to reduce or inhibit incoming dirty air exiting the inlet port **164** from travelling directly from the chamber air inlet **160** to the chamber air outlet **162**, which may otherwise result in particulate matter clogging or building up on the porous separator **170**.

[0150] The air impervious cover **176** may induce directional changes in the airflow as it deflects off of the air impervious cover **176** and optionally the sidewall **152** of the air treatment chamber **120** as the air travels to the chamber air outlet **162**. Accordingly, turbulence in the airflow may be created which may result in improved disentrainment of particulate matter. For example, the air impervious cover **176** may cause the formation of eddies in the airflow as it passes by (e.g., under) the air impervious cover **176**.

[0151] Referring now to FIGS. **20A** to **20C**, the hand vacuum **100** includes an air impervious cover **176** positioned at the rear end **150** of the air treatment chamber **120**. As shown, the air impervious cover **176** is spaced from the chamber air outlet **162** and overlies the chamber air outlet **162**. Accordingly, the conduit axis **142** of the air inlet conduit **140** intersects the air impervious cover **176**.

[0152] As shown, the air impervious cover **176** has a front side **252**, a rear side **254**, a radial outer edge **256**, and a free edge **258**. As exemplified, the radial outer edge **256** of the air impervious cover **176** is provided on the inner surface of the sidewall **152** such that the air impervious cover **176** follows the shape of the air treatment chamber **120**. The free edge **258** of air impervious cover **176** has a first lateral end **260a** and a second lateral end **260b** and is unattached to any portion of the sidewall **152** between the first and second ends **260a**, **260b**. The air impervious cover **176** may extend from the sidewall **152** in a direction generally transverse to the longitudinal axis **158** of the air treatment chamber **120** and, as exemplified, may be generally semi-circular in a plane transverse

to the longitudinal axis **158**.

[0153] It will be appreciated that, similar to baffle **174**, any other shape may be possible.

Accordingly, the radial outer edge **256** of the air impervious cover **176** may be partially or fully spaced from the inner surface of the sidewall **152**.

[0154] Further, similar to the baffle **174**, the free edge **258** may have any shape between the first and second lateral ends **260a**, **260b**. For example, the free edge **258** may be linear such as shown or may be shaped to further influence the airflow path **126** as it diverts around the air impervious cover **176**. For example, the air impervious cover **176** may have one or more recesses that extend radially outwardly and/or one or more protrusions that extend radially inwardly, which may split the airflow path **126** into two or more airflow paths **126** as described previously with respect to the baffle plate **174**. Alternatively, or in addition to the recesses and/or protrusions provided on the free edge **258** of the air impervious cover **176**, the front side **252** of the air impervious cover **176** may include one or more forwardly extending portions to split the airflow as it impacts on the front side **252** as described previously with respect to the baffle plate **174**.

[0155] The free edge **258** of the air impervious cover **176** may be positioned at any elevation within air treatment chamber **120** from a facing portion of the sidewall **152**. The positioning of the free edge **258** may influence the degree of directional change in the airflow path **126** as it diverts around the air impervious cover **176** while travelling from the chamber air inlet **160** to the chamber air outlet **162**, which may influence the amount of turbulence induced in the airflow as described previously. For example, when the hand vacuum **100** is oriented with the upper end **112** above the lower end **114**, the free edge **258** may be below or aligned with a lower end of the chamber air inlet **160** and/or below or aligned with an upper end of the chamber air outlet **162**. Optionally, the free edge **258** may be below or aligned with a lower end of the chamber air outlet **162**. In the illustrated example, the free edge **242** is below the lower end of the chamber air inlet **160** and the lower end of the chamber air outlet **162**. Accordingly, in use as exemplified in FIG. 21C, the airflow path **126** through the air treatment chamber **120** includes a portion that extends upwardly from below the air impervious cover **176** (e.g., generally radially inwardly) at a location rearward of the air impervious cover **176** to travel from below the air impervious cover **176** to the chamber air outlet **162**.

[0156] In embodiments wherein sidewall **152** of the air treatment chamber **120** includes the openable portion **182**, the free edge **258** of the air impervious cover **176** may be positioned at an elevation above (e.g., adjacent) an upper end of the openable portion **182**. Similarly, in embodiments wherein front end wall **166** of the air treatment chamber **120** includes the openable door **184**, the free edge **258** of the air impervious cover **176** may be positioned at an elevation above or aligned with an upper end of the openable door **184**. In such embodiments, the described positioning of the air impervious cover **176** may provide an unobstructed path for emptying the volume **180** of the air treatment chamber **120**.

[0157] It will be appreciated that, if the chamber air outlet includes a porous separator **170**, then the air impervious cover **176** may be positioned overlying part or all of the porous separator **170** and may be axially located at the axial inner end of the porous separator **170** or axially inward thereof. For example, the air impervious cover **176** may be positioned adjacent the porous separator **170** of FIGS. 6, 10A, 11A, or 12A. In the example illustrated in FIGS. 20B and 20C, the air impervious cover **176** is positioned overlying a flat porous separator **170**, similar to that of FIG. 11A, which is vertically oriented over to chamber air outlet **162**. In this embodiment, the flat porous separator **170** has a vertical extent that is about equal to the radius of the volume **180** and the air impervious cover **176** overlies all of the porous separator **170**.

[0158] In alternate embodiments, the air impervious cover **176** may be integrated with the porous separator **170**. For example, as shown in FIGS. 20D and 20E, a flat porous separator **170** extends rearwardly (horizontally as exemplified) from a lower end of the air impervious cover **176**. In the illustrated example, the porous separator **170** extends from the rear end wall **168** of the air

treatment chamber **120** below the chamber air outlet **162** to the rear side **254** of the air impervious cover **176** proximate the free edge **258**.

[0159] As exemplified in FIGS. **21A** to **21C**, the air impervious cover **176** may be positioned closer to the chamber air inlet **160** to function more similarly to the baffle plate **174** as described herein. In such embodiments, the porous separator **170** may extend a greater distance from the rear end wall **168** of the air treatment chamber **120** to the rear side **254** of the air impervious cover **176**, providing more surface area for the rigid screen **226**.

Bleed Air Inlet Port Provided Through the Sidewall of the Air Treatment Chamber

[0160] A hand vacuum cleaner using any one or more aspects discussed herein, including one or more of the other aspects set out herein including one or more of the porous separator positioned at the chamber air outlet, the non-return valve flap that closes an inlet port of the chamber air inlet, the baffle plate positioned in the air treatment chamber rearward of the chamber air inlet, the air impervious cover positioned at the rear end of the air treatment chamber and overlying the porous separator, and the axially extending rib provided on an inner surface of the sidewall of the air treatment chamber, may have the bleed air inlet port provided through the sidewall of the air treatment chamber.

[0161] Referring back to FIGS. **9A** to **9C**, in the illustrated example, the sidewall **152** of the air treatment chamber **120** has an openable bleed air inlet **262** provided through a portion of the sidewall **152**, which may be the openable portion **182** of the sidewall **152** if an openable portion **182** is provided. While the bleed air inlet **262** is shown through the sidewall **152** at the lower end **154** of the air treatment chamber **120**, it will be appreciated that the bleed air inlet **262** may be provided at any other location suitable for providing relief/assisting airflow. The relief/assisting airflow provided through the bleed air inlet **262** may advantageously enable the use of a smaller suction motor in the docking station **186** and/or prevent burnout of the suction motor therein.

[0162] The bleed air inlet **262** may include a moveable seal **264**, which may be movable (e.g., slidably or rotatably) between a closed position in which the bleed air inlet **262** is closed (see e.g., FIG. **9B**) and an open position (see e.g., FIG. **9A**) in which the bleed air inlet **262** is open. The moveable seal **264** may be provided on the inner surface of the sidewall **152** or provided on the outer surface of the sidewall **152**. In the illustrated example, the moveable seal **264** is slidably provided on the external surface of the sidewall **152**. The moveable seal **264** may be lockable in the closed position, such as by a latch or other suitable mechanism. It will be appreciated that the moveable seal **264** may move in any direction to open the bleed air inlet **262** and may be secured in the closed position in any manner.

[0163] As described previously, when the hand vacuum **100** is docked to the docking station **186** (see e.g., FIG. **9A**), the openable door **184** may, manually or automatically, move from the closed position (see e.g., FIG. **9B**) to the open position (see e.g., FIG. **9C**). When in the openable door **184** is in open position and the hand vacuum **100** is docked to the docking station **186**, the volume **180** may be in fluid communication with the suction motor (not shown) of the station base **188** via the upright conduit **190**. Similarly, the moveable seal **264** of the bleed air inlet **262** may, manually or automatically, move from the closed position (see e.g.,

[0164] FIG. **9B**) to the open position (see e.g., FIG. **9C**) upon or subsequent to docking with the docking station **186** or, in some embodiments, upon detection of the need for relief airflow to the suction motor of the base station **188**. When in the moveable seal **264** is in the open position and the hand vacuum **100** is docked to the docking station **186**, the suction motor of the station base **188** may draw air inwardly through the bleed air inlet **262** to assist in emptying the volume **180** of the air treatment chamber **120**. For example, once the openable door **184** and moveable seal **264** are in their respective open positions, the suction motor in the station base **188** may, during an emptying cycle, manually or automatically, activate and generate suction to draw the contents of the volume **180** within the air treatment chamber **120** through the opening of the openable door **184**, through the upright conduit **190**, and into a collection chamber (not shown) within the station

base **188**. It will accordingly be appreciated that, when the hand vacuum **100** is docked, the bleed air inlet **262** is in communication with the ambient and, as such, is optionally exterior to a docking cradle of the docking station.

[0165] It will be appreciated that, during an emptying cycle, air may be drawn in a revise flow from the clean air outlet **124**, through a post-motor filter if provided, past the suction motor, through a pre-motor filter if provided and through the chamber air outlet **162** into the volume **180**.

Accordingly, providing a bleed air inlet **262** may enhance the air flow through the volume **180** during an emptying cycle.

[0166] It will be appreciated that, in some embodiments, the non-return valve **172** may be opened to enable air to enter the volume **180** during an emptying cycle. In such a case, the suction motor of the station base **188** may need to generate sufficient suction to move the non-return valve flap **230** from the closed position to the open position, whereby air may then be drawn through the air inlet conduit **140** and, if attached, through the wand **106** and surface cleaning head **104**. Accordingly, without the bleed air inlet **262**, a larger, higher power suction motor may therefore be required to generate sufficient suction force to open the non-return valve **172** and to draw air through the cumulative distance of the path through the upright conduit **190**, the air treatment chamber **120**, the air inlet conduit **140** and, if attached, the wand **106** and surface cleaning head **104**.

[0167] When the hand vacuum **100** is docked to the docking station **186**, collected particulate matter that has fallen toward the front end **148** of the air treatment chamber **120** may prevent or inhibit the non-return valve flap **230** from opening enough to provide sufficient airflow to the suction motor of the station base **188**. Accordingly, in some embodiments, if the bleed air inlet **262** is provided, then the moveable seal **264** may automatically move to the open position to open the bleed air inlet **262** and provide relief airflow to the suction motor if there is insufficient airflow. In such embodiments, the bleed air inlet **262** may thus help prevent burnout of the suction motor in the station base **188**.

[0168] In alternate embodiments, the bleed air inlet **262** may serve as the primary source or a source of airflow to the suction motor of the station base **188**. Using the bleed air inlet **262** may reduce the total distance from which the suction motor must draw in air, and thus reduce the power requirements of the suction motor of the station base **188**. Optionally, this may also eliminate the suction force requirement for opening the non-return valve **172**.

[0169] While the above description describes features of example embodiments, it will be appreciated that some features and/or functions of the described embodiments are susceptible to modification without departing from the spirit and principles of operation of the described embodiments. For example, the various characteristics which are described by means of the represented embodiments or examples may be selectively combined with each other. Accordingly, what has been described above is intended to be illustrative of the claimed concept and non-limiting. It will be understood by persons skilled in the art that other variants and modifications may be made without departing from the scope of the invention as defined in the claims appended hereto. The scope of the claims should not be limited by the preferred embodiments and examples, but should be given the broadest interpretation consistent with the description as a whole.

[0170] This specification also includes the subject matter of the following clause sets:

Clause Set A

[0171] 1. A hand vacuum cleaner comprising: [0172] (a) an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet, wherein a suction motor is positioned in the air flow path; [0173] (b) a non-cyclonic air treatment chamber in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet, a sidewall extending between the front and rear ends of the air treatment chamber and a longitudinal axis extending between the front and rear ends of the air treatment chamber, wherein the chamber air inlet has an inlet axis that extends

rearwardly; and, [0174] (c) a baffle plate positioned in the air treatment chamber rearward of the chamber air inlet and the inlet axis intersects the baffle plate.

[0175] 2. The hand vacuum cleaner of clause 1 wherein the baffle plate is generally semi-circular in a plane transverse to the longitudinal axis.

[0176] 3. The hand vacuum cleaner of clause 1 wherein the baffle plate is located along a portion of an inner surface of the sidewall.

[0177] 4. The hand vacuum cleaner of clause 1 wherein a radial outer edge of the baffle plate is provided on the inner surface of the sidewall and the baffle plate has a free edge and the free edge is planar.

[0178] 5. The hand vacuum cleaner of clause 1 wherein a radial outer edge of the baffle plate is provided on the inner surface of the sidewall and the baffle plate has a free edge having a recess that extends radially outwardly.

[0179] 6. The hand vacuum cleaner of clause 1 wherein a radial outer edge of the baffle plate is provided on the inner surface of the sidewall and the baffle plate has a free edge having two spaced apart recesses that extends radially outwardly.

[0180] 7. The hand vacuum cleaner of clause 1 wherein a radial outer edge of the baffle plate is provided on the inner surface of the sidewall and the baffle plate has a free edge having a protrusion that extends radially inwardly.

[0181] 8. The hand vacuum cleaner of clause 1 wherein the sidewall has an axially extending rib provided on an inner surface of the sidewall rearward of the baffle plate.

[0182] 9. The hand vacuum cleaner of clause 1 wherein the baffle plate extends generally transverse to the longitudinal axis.

[0183] 10. The hand vacuum cleaner of clause 1 wherein the sidewall has an openable portion, a rear side of the baffle plate has a radial outer end and a radial inner end, the radial inner end is located closer to the openable portion than the radial outer end and the radial outer end is located further axially inwardly than the radial inner end.

[0184] 11. The hand vacuum cleaner of clause 1 wherein the sidewall has an openable portion and a rear side of the baffle plate is shaped such that, when the dirty air inlet faces downwardly, the rear side directs dirt towards the openable portion.

[0185] 12. The hand vacuum cleaner of clause 1 further comprising a non-return valve flap associated with the chamber air inlet.

[0186] 13. The hand vacuum cleaner of clause 1 wherein a radial outer edge of the baffle plate is provided at or adjacent an inner surface of the sidewall, the baffle plate has a free edge having first and second laterally spaced apart sides and a front side of the baffle plate has a portion centrally located between the first and second laterally spaced apart sides that extends axially forwardly.

[0187] 14. The hand vacuum cleaner of clause 1 wherein the sidewall has an openable portion and the openable portion has an openable inlet.

[0188] 15. A hand vacuum cleaner comprising: [0189] (d) an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet, wherein a suction motor is positioned in the air flow path; [0190] (e) a non-cyclonic air treatment chamber in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet and a longitudinal axis extending between the front and rear ends of the air treatment chamber; and, [0191] (f) a baffle plate positioned in the air treatment chamber rearward of the chamber air inlet overlying an outlet port of the chamber air inlet.

[0192] 16. The hand vacuum cleaner of clause 15 wherein the baffle plate is generally semi-circular in a plane transverse to the longitudinal axis and is located along a portion of an inner surface of the sidewall.

[0193] 17. The hand vacuum cleaner of clause 15 wherein the baffle plate extends generally

transverse to the longitudinal axis.

[0194] 18. The hand vacuum cleaner of clause 15 wherein the sidewall has an openable portion and a rear side of the baffle plate is shaped such that, when the dirty air inlet faces downwardly, the rear side directs dirt towards the openable portion.

[0195] 19. The hand vacuum cleaner of clause 15 further comprising a non-return valve flap associated with the chamber air inlet.

[0196] 20. The hand vacuum cleaner of clause 15 wherein a radial outer edge of the baffle plate is provided at or adjacent an inner surface of the sidewall, the baffle plate has a free edge having first and second laterally spaced apart sides and a front side of the baffle plate has a portion centrally located between the first and second laterally spaced apart sides that extends axially forwardly.

Clause Set B

[0197] 1. A hand vacuum cleaner comprising: [0198] (a) an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet, wherein a suction motor is positioned in the air flow path; [0199] (b) a non-cyclonic air treatment chamber in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet, a sidewall extending between the front and rear ends of the air treatment chamber and a longitudinal axis extending between the front and rear ends of the air treatment chamber, wherein the chamber air outlet comprises a porous separator; and, [0200] (c) an air impervious cover positioned at the rear end of the air treatment chamber and overlying the porous separator whereby, an air flow path through the air treatment chamber comprises a portion that extends generally radially inwardly at a location rearward of the air impervious cover.

[0201] 2. The hand vacuum cleaner of clause 1 wherein the air impervious cover is positioned adjacent the porous separator.

[0202] 3. The hand vacuum cleaner of clause 1 wherein the air impervious cover is generally semi-circular in a plane transverse to the longitudinal axis.

[0203] 4. The hand vacuum cleaner of clause 1 wherein the air impervious cover is located along a portion of an inner surface of the sidewall.

[0204] 5. The hand vacuum cleaner of clause 1 wherein a radial outer edge of the air impervious cover is provided on an inner surface of the sidewall and the air impervious cover has a free edge.

[0205] 6. The hand vacuum cleaner of clause 1 wherein the porous separator comprises a rigid screen.

[0206] 7. The hand vacuum cleaner of clause 1 wherein the porous separator is generally planar.

[0207] 8. The hand vacuum cleaner of clause 1 wherein the porous separator is generally planar and extends generally perpendicular to the longitudinal axis.

Claims

1. A hand vacuum cleaner comprising: (a) an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet, wherein a suction motor is positioned in the air flow path; (b) a handle provided on a lower side of the hand vacuum cleaner; (c) a non-cyclonic air treatment chamber in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet and a longitudinal axis extending between the front and rear ends of the air treatment chamber; and, (d) a non-return valve flap that closes an inlet port of the chamber air inlet, the non-return valve flap having an upper end that is mounted to the air treatment chamber and a lower end, wherein the non-return valve flap is moveable from a closed position in which it closes the inlet port to an open position in which the lower end of the non-return valve flap rotates inwardly and upwardly whereby the inlet port is

opened.

2. The hand vacuum cleaner of claim 1 wherein, in the open position, the non-return valve flap rotates open less than 90°, whereby air exiting the inlet port is directed in a non-cyclonic flow towards a lower side of the air treatment chamber.
3. The hand vacuum cleaner of claim 1 wherein the air treatment chamber comprises a sidewall extending between the front and rear ends of the air treatment chamber, the sidewall has an axially extending rib provided on an inner surface of the sidewall.
4. The hand vacuum cleaner of claim 1 wherein the inlet port is provided on an uppermost end of the air treatment chamber.
5. The hand vacuum cleaner of claim 4 wherein, in the open position, the non-return valve flap rotates open less than 90°, whereby air exiting the inlet port is directed towards a lower side of the air treatment chamber.
6. The hand vacuum cleaner of claim 1 wherein the chamber air inlet has an inlet axis that extends rearwardly, the hand vacuum cleaner further comprises a baffle plate positioned in the air treatment chamber rearward of the non-return valve flap when the non-return valve flap is in the open position and the inlet axis intersects the baffle plate.
7. The hand vacuum cleaner of claim 1 wherein the chamber air outlet comprises a rigid screen which has a generally frusto-conical section.
8. The hand vacuum cleaner of claim 1 wherein the air treatment chamber comprises a sidewall extending between the front and rear ends of the air treatment chamber, the rear end of the air treatment chamber comprises a rear wall, the chamber air outlet comprises a rigid screen which is provided on the rear wall spaced radially inwardly from the sidewall.
9. A hand vacuum cleaner comprising: (a) an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet, wherein a suction motor is positioned in the air flow path and the dirty air inlet is provided at an upper end of the hand vacuum cleaner; (b) a non-cyclonic air treatment chamber in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet and a longitudinal axis extending between the front and rear ends of the air treatment chamber; and, (c) a non-return valve flap that closes an inlet port of the chamber air inlet, the non-return valve flap having an upper end that is mounted to the air treatment chamber and a lower end, wherein the non-return valve flap is moveable from a closed position in which it closes the inlet port to an open position in which the lower end of the non-return valve flap rotates inwardly and upwardly whereby the inlet port is opened.
10. The hand vacuum cleaner of claim 9 wherein, in the open position, the non-return valve flap rotates open less than 90°, whereby air exiting the inlet port is directed towards a lower side of the air treatment chamber.
11. The hand vacuum cleaner of claim 9 wherein the air treatment chamber comprises a sidewall extending between the front and rear ends of the air treatment chamber, the sidewall has an axially extending rib provided on an inner surface of the sidewall.
12. The hand vacuum cleaner of claim 9 wherein the inlet port is provided on an uppermost end of the air treatment chamber.
13. The hand vacuum cleaner of claim 12 wherein, in the open position, the non-return valve flap rotates open less than 90°, whereby air exiting the inlet port is directed towards a lower side of the air treatment chamber.
14. The hand vacuum cleaner of claim 9 wherein the chamber air inlet has an inlet axis that extends rearwardly, the hand vacuum cleaner further comprises a baffle plate positioned in the air treatment chamber rearward of the non-return valve flap when the non-return valve flap is in the open position and the inlet axis intersects the baffle plate.
15. The hand vacuum cleaner of claim 14 wherein the chamber air outlet comprises a rigid screen

which has a generally frusto-conical section.

16. The hand vacuum cleaner of claim 14 wherein the air treatment chamber comprises a sidewall extending between the front and rear ends of the air treatment chamber, the rear end of the air treatment chamber comprises a rear wall, the chamber air outlet comprises a rigid screen which is provided on the rear wall spaced radially inwardly from the sidewall.

17. A hand vacuum cleaner comprising: (a) an air flow path from a dirty air inlet provided at a front end of the hand vacuum cleaner to a clean air outlet that is positioned rearward of the dirty air inlet, wherein a suction motor is positioned in the air flow path; (b) a non-cyclonic air treatment chamber in the air flow path downstream from the dirty air inlet and upstream from the suction motor, the air treatment chamber having a front end having a chamber air inlet, a rear end having a chamber air outlet, a sidewall extending between the front and rear ends of the air treatment chamber and a longitudinal axis extending between the front and rear ends of the air treatment chamber; and, (c) a non-return valve flap that closes an inlet port of the chamber air inlet, the non-return valve flap having a first end that is mounted to the air treatment chamber at a location adjacent a first portion of an inner surface of the sidewall and an opposed end, wherein the non-return valve flap is moveable from a closed position in which it closes the inlet port to an open position in which the opposed end of the non-return valve flap rotates inwardly and towards the sidewall whereby the inlet port is opened and air exiting the inlet port is directed across a volume of the non-cyclonic air treatment chamber towards a second portion of the sidewall that is opposed to the first portion in a direction generally transverse to the longitudinal axis.

18. The hand vacuum cleaner of claim 17 wherein, in the open position, the non-return valve flap rotates open less than 90° , whereby air exiting the inlet port is directed towards the second portion of the sidewall.

19. The hand vacuum cleaner of claim 17 wherein the sidewall has an axially extending rib provided on an inner surface of the sidewall.

20. The hand vacuum cleaner of claim 17 wherein the chamber air inlet has an inlet axis that extends rearwardly, the hand vacuum cleaner further comprises a baffle plate positioned in the air treatment chamber rearward of the non-return valve flap when the non-return valve flap is in the open position and the inlet axis intersects the baffle plate.
